SteadyStud — Magnet Specs + Assembly Notes (Draft)

Goal: magnet configuration strong enough to reliably "grab" drywall screws through paint/drywall, while being safe to handle and manufacturable.

Recommended v1 (simple + low risk)

Option 1 — Single plastic-coated cylindrical magnet (off-the-shelf)

- Type: Neodymium magnet in plastic/rubber coating
- Suggested size starting point: **1/2" dia × 5/8" length** (or similar)
- Suggested grades to quote/source: **N42** and **N52** (N52 stronger; can be overkill)
- Why: easy sourcing, consistent, coating reduces chipping/corrosion and improves feel

Mounting - Create a pocket in the aluminum body sized for a light press-fit + adhesive. - Adhesive: 2-part epoxy (preferred) for permanent bond.

Higher-force v1.1 (still simple)

Option 2 — Dual disc magnets (stacked)

- Type: 2× disc magnets stacked in a pocket (increases pull force)
- Suggested size starting point: 2× **1/2" dia × 1/8" thick** discs
- Grade: **N52**
- Pros: strong, easy to iterate by swapping discs
- Cons: needs good pocket design; discs can snap/chip if uncoated

Mounting - Pocket + epoxy. Consider a thin polymer cap/insert to reduce chip risk.

Premium variant (future)

Option 3 — Spherical magnet ("Stud Ball" style)

- Type: N52 sphere, **3/4"–1"**
- Pros: maximum detection range / premium marketing angle
- Cons: safety/handling risk, higher cost, different industrial design

Sourcing notes

- Prototype: buy small quantities from reputable magnet suppliers (US) to validate strength quickly.
- Production: direct factory sourcing once final size/grade is locked.

Assembly recommendation (v1)

1) Machine aluminum pocket
2) Clean/degrease pocket and magnet surface
3) Apply epoxy (controlled amount)
4) Insert magnet (check polarity orientation if relevant)
5) Cure under simple jig/fixture

QA / Testing

- Incoming inspection: dimensions + coating integrity
- Pull-force sanity checks: repeatable "grab" test on standard drywall screw behind sample drywall
- Drop test: confirm magnet stays seated and coating doesn't crack

Questions to vendors (if they're involved in magnet integration)

- Can you machine the pocket to hold this magnet tolerance reliably?
- Can you add a retention feature (lip/cap) if epoxy-only is not desired?
- What's your preferred adhesive for aluminum + coated magnet?